

## 1. Create a CloudFormation template that does the following:

1. Create a VPC with CIDR block passed as a parameter.
2. Create one Subnet inside that VPC.
3. Launch one EC2 instance inside the subnet.
4. Create one S3 bucket with name passed as parameter.
5. Use Metadata to group parameters into: Network Configuration, EC2 Configuration, S3 Configuration
6. Add Tags to all resources with: Key: Project  
Value: Banking

Ans:

```
! Q1.yaml x
home > piyush > Desktop > ! Q1.yaml
1  AWSTemplateFormatVersion: '2010-09-09'
2  Description: 'AWS CloudFormation Template: Question 1 YAML File Template'
3
4  Metadata:
5    AWS::CloudFormation::Interface:
6      ParameterGroups:
7        - Label:
8            default: "Network Configuration"
9          Parameters:
10            - VPCCIDR
11            - SubnetCIDR
12
13        - Label:
14            default: "EC2 Configuration"
15          Parameters:
16            - InstanceType
17            - KeyName
18            - AMIID
19
20        - Label:
21            default: "S3 Configuration"
22          Parameters:
23            - BucketName
24
25
26  Parameters:
27
28    VPCCIDR:
29      Type: String
30      Description: CIDR block for the VPC
31      Default: 10.0.0.0/16
32
33    SubnetCIDR:
34      Type: String
35      Description: CIDR block for the Subnet
36      Default: 10.0.1.0/24
37
38    InstanceType:
39      Type: String
40      Description: EC2 Instance Type
41      Default: t3.micro
42      AllowedValues:
43        - t3.micro
44        - t3.small
45        - t3.medium
46
47    KeyName:
48      Type: AWS::EC2::KeyPair::KeyName
49      Description: Name of an existing EC2 KeyPair to enable SSH access
50
```

home &gt; piyush &gt; Desktop &gt; ! Q1.yaml

```
26 Parameters:
47   KeyName:
50
51   AMIID:
52     Type: AWS::EC2::Image::Id
53     Description: AMI ID for the EC2 instance
54
55   BucketName:
56     Type: String
57     Description: Name of the S3 bucket to be created
58
59
60 Resources:
61
62   MyVPC:
63     Type: AWS::EC2::VPC
64     Properties:
65       CidrBlock: !Ref VPCCIDR
66       Tags:
67         - Key: Name
68           Value: Banking-VPC
69         - Key: Project
70           Value: Banking
71
72   MySubnet:
73     Type: AWS::EC2::Subnet
74     Properties:
75       VpcId: !Ref MyVPC
76       CidrBlock: !Ref SubnetCIDR
77       MapPublicIpOnLaunch: true
78       Tags:
79         - Key: Name
80           Value: Banking-Subnet
81         - Key: Project
82           Value: Banking
83
84   InternetGateway:
85     Type: AWS::EC2::InternetGateway
86     Properties:
87       Tags:
88         - Key: Project
89           Value: Banking
90
91   AttachGateway:
92     Type: AWS::EC2::VPCGatewayAttachment
93     Properties:
94       InternetGatewayId: !Ref InternetGateway
95       VpcId: !Ref MyVPC
96
97   RouteTable:
```

```
! Q1.yaml x
home > piyush > Desktop > ! Q1.yaml
60 Resources:
91   AttachGateway:
96
97   RouteTable:
98     Type: AWS::EC2::RouteTable
99     Properties:
100       VpcId: !Ref MyVPC
101       Tags:
102         - Key: Project
103           Value: Banking
104
105   DefaultRoute:
106     Type: AWS::EC2::Route
107     DependsOn: AttachGateway
108     Properties:
109       RouteTableId: !Ref RouteTable
110       DestinationCidrBlock: 0.0.0.0/0
111       GatewayId: !Ref InternetGateway
112
113   SubnetRouteTableAssociation:
114     Type: AWS::EC2::SubnetRouteTableAssociation
115     Properties:
116       SubnetId: !Ref MySubnet
117       RouteTableId: !Ref RouteTable
118
119   InstanceSecurityGroup:
120     Type: AWS::EC2::SecurityGroup
121     Properties:
122       GroupDescription: Allow SSH access
123       VpcId: !Ref MyVPC
124       SecurityGroupIngress:
125         - IpProtocol: tcp
126           FromPort: 22
127           ToPort: 22
128           CidrIp: 0.0.0.0/0
129       Tags:
130         - Key: Project
131           Value: Banking
132
133   MyEC2Instance:
134     Type: AWS::EC2::Instance
135     Properties:
136       InstanceType: !Ref InstanceType
137       KeyName: !Ref KeyName
138       ImageId: !Ref AMIID
139       SubnetId: !Ref MySubnet
140       SecurityGroupIds:
141         - !Ref InstanceSecurityGroup
142       Tags:
143         - Key: Name
```

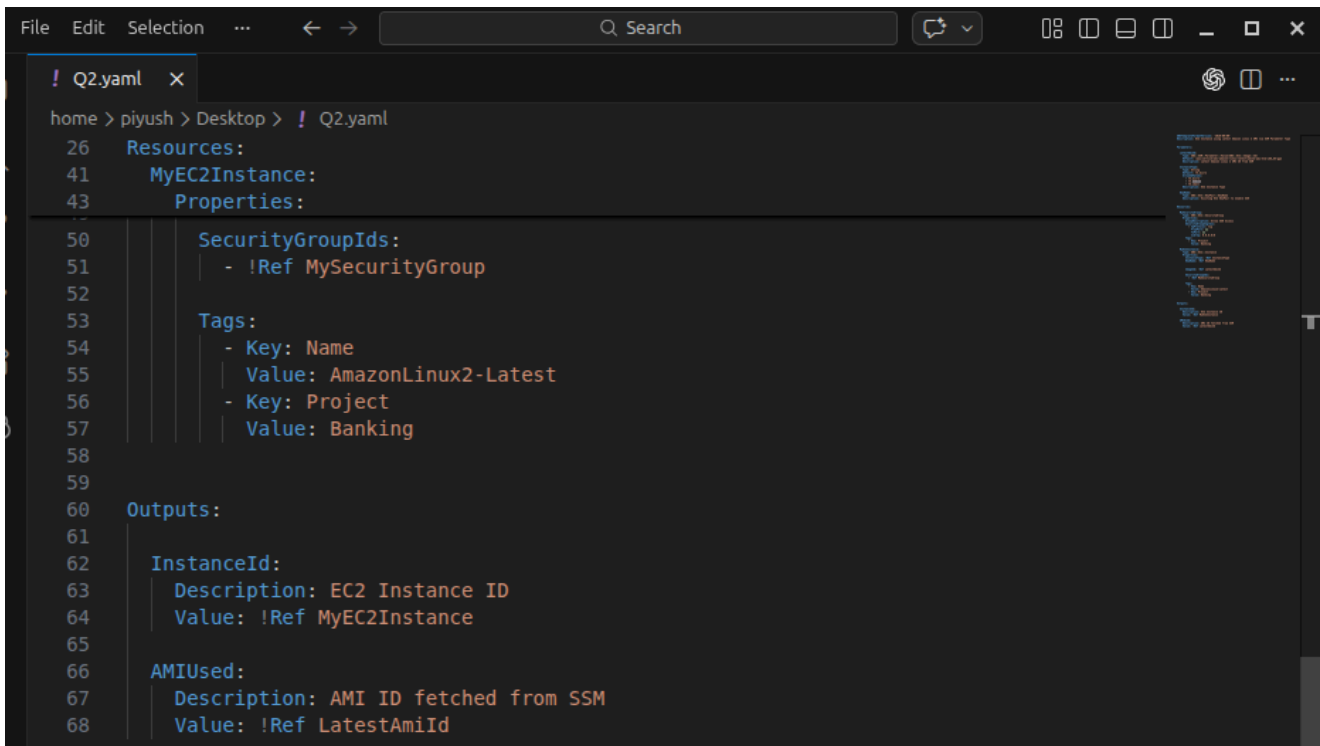
```
! Q1.yaml x
home > piyush > Desktop > ! Q1.yaml
60   Resources:
133     MyEC2Instance:
135       Properties:
142         Tags:
145           - Key: Project
147
148     MyS3Bucket:
149       Type: AWS::S3::Bucket
150       Properties:
151         BucketName: !Ref BucketName
152         Tags:
153           - Key: Project
154             Value: Banking
155
156   Outputs:
157     VPCId:
158       Description: VPC ID
159       Value: !Ref MyVPC
160
161     SubnetId:
162       Description: Subnet ID
163       Value: !Ref MySubnet
164
165     InstanceId:
166       Description: EC2 Instance ID
167       Value: !Ref MyEC2Instance
168
169     BucketName:
170       Description: S3 Bucket Name
171       Value: !Ref MyS3Bucket
172
173
```

**2. Create a CloudFormation template that automatically fetches the latest Amazon Linux 2 AMI ID using AWS Systems Manager (SSM) Parameter Store instead of hardcoding the AMI.**

Ans:

```
File Edit Selection ... ← → Q Search [Icons]
! Q2.yaml x [Icons] ...

home > piyush > Desktop > ! Q2.yaml
1  AWSTemplateFormatVersion: '2010-09-09'
2  Description: EC2 Instance using Latest Amazon Linux 2 AMI via SSM Parameter Type
3
4
5  Parameters:
6
7    LatestAmiId:
8      Type: AWS::SSM::Parameter::Value<AWS::EC2::Image::Id>
9      Default: /aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2
10     Description: Latest Amazon Linux 2 AMI ID from SSM
11
12    InstanceType:
13      Type: String
14      Default: t3.micro
15      AllowedValues:
16        - t3.micro
17        - t3.medium
18        - t3.small
19      Description: EC2 Instance Type
20
21    KeyName:
22      Type: AWS::EC2::KeyPair::KeyName
23      Description: Existing EC2 KeyPair to enable SSH
24
25
26  Resources:
27
28    MySecurityGroup:
29      Type: AWS::EC2::SecurityGroup
30      Properties:
31        GroupDescription: Allow SSH Access
32        SecurityGroupIngress:
33          - IpProtocol: tcp
34            FromPort: 22
35            ToPort: 22
36            CidrIp: 0.0.0.0/0
37        Tags:
38          - Key: Project
39            Value: Banking
40
41    MyEC2Instance:
42      Type: AWS::EC2::Instance
43      Properties:
44        InstanceType: !Ref InstanceType
45        KeyName: !Ref KeyName
46
47
48        ImageId: !Ref LatestAmiId
```



```
File Edit Selection ... ← → Search [Icons]
! Q2.yaml x
home > piyush > Desktop > ! Q2.yaml
26 Resources:
41   MyEC2Instance:
43     Properties:
50       SecurityGroupIds:
51         - !Ref MySecurityGroup
52
53       Tags:
54         - Key: Name
55           Value: AmazonLinux2-Latest
56         - Key: Project
57           Value: Banking
58
59
60 Outputs:
61
62   InstanceId:
63     Description: EC2 Instance ID
64     Value: !Ref MyEC2Instance
65
66   AMIUsed:
67     Description: AMI ID fetched from SSM
68     Value: !Ref LatestAmiId
```

### 3. Your company wants to deploy a highly available three-tier application in the eu-west-2 region.

Requirements:

- The architecture must include Web, Application, and Database tiers.
- Web servers must auto-scale between 1–2 instances.
- Application servers must auto-scale between 1–2 instances.
- Servers must be deployed across 2 Availability Zones.
- Traffic must be routed through an Application Load Balancer to the Web tier.
- The Web tier must communicate only with the Application tier.
- The Application tier must communicate only with the Database tier.
- The Database tier must use Amazon RDS with Multi-AZ enabled
- EC2 instances must install application configuration automatically during launch (using User Data).
- The load balancer must distribute traffic evenly across healthy instances.

Ans:

CloudFormation > Stacks > tier3-web-app

Stacks (1)

Filter status

Search by stack name

Active

View nested

Stacks

tier3-web-app

2036-02-24 17:19:54 UTC+0530

UPDATE\_COMPLETE

tier3-web-app

DeleteUpdate stackStack actionsCreate stack

Stack infoEventsResourcesOutputsParametersTemplateChange setsGit sync

Outputs (7)

Search outputs

| Key                  | Value                                                                                         | Description                                                                                           | Export name |
|----------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------|
| AppBucketName        | tier3-web-app-appbucket-8xb7pkhietb                                                           | S3 bucket containing app PHP files (upload index.php, add-user.php, users.php from Resources/webapp/) | -           |
| AutoScalingGroupName | tier3-app-asg                                                                                 | Auto Scaling Group Name                                                                               | -           |
| LoadBalancerDNS      | tier3-app-alb-1508895037-us-east-1.elb.amazonaws.com                                          | DNS name of the Application Load Balancer                                                             | -           |
| LoadBalancerURL      | http://tier3-app-alb-1508895037-us-east-1.elb.amazonaws.com                                   | URL of the Application Load Balancer                                                                  | -           |
| RDSEndpoint          | tier3-app-db-c43omjyq080.us-east-1.rds.amazonaws.com                                          | RDS Database Endpoint                                                                                 | -           |
| TargetGroupARN       | arn:aws:elasticloadbalancing:us-east-1:596059883112:targetgroup/tier3-app-tg/4db91c1d520d5d10 | Target Group ARN                                                                                      | -           |
| VPCId                | vpc-041d4a9e5de5dea2c                                                                         | VPC ID                                                                                                | -           |

Resource mapCIDRsFlow logsTagsIntegrations

Resource map

Show all details

VPC

Your AWS virtual network

3tier-vpc

10.0.0.0/16

No IPv6

Subnets (6)

Subnets within this VPC

us-east-1a

3tier-alb-publicA

10.0.1.0/24

No IPv6

3tier-db-privateA

10.0.21.0/24

No IPv6

3tier-app-privateA

10.0.11.0/24

No IPv6

us-east-1b

3tier-alb-publicB

10.0.2.0/24

No IPv6

3tier-db-privateB

10.0.22.0/24

No IPv6

3tier-app-privateB

10.0.12.0/24

No IPv6

Route tables (4)

Route network traffic to resources

rtb-03811e1991472becf

No subnet associations

1 route including local

3tier-db-private-rt

2 subnet associations

1 route including local

3tier-app-private-rt

2 subnet associations

2 routes including local

3tier-public-rt

2 subnet associations

2 routes including local

Network Connections (2)

Connections to other networks

3tier-igw

Internet routes to 2 public subnets

2 private subnets route to the Internet

3tier-natgw

Public NAT gateway

1 ENI with 1 EIP

Security Groups (3/5) 

Find security groups by attribute or tag

ActionsExport security groups to CSVCreate security group

|                                     | Name         | Security group ID    | Security group name                     | VPC ID                | Description                                | Owner        |
|-------------------------------------|--------------|----------------------|-----------------------------------------|-----------------------|--------------------------------------------|--------------|
| <input checked="" type="checkbox"/> | 3tier-rds-sg | sg-05832017a4c8d919d | tier3-web-app-RDSSecurityGroup-qh5aj... | yvc-041d4a9e5de5dea2c | Security group for RDS database            | 596059883112 |
| <input checked="" type="checkbox"/> | 3tier-app-sg | sg-09c143a7fd0ac1f96 | tier3-web-app-AppSecurityGroup-5MW...   | yvc-041d4a9e5de5dea2c | Security group for application servers     | 596059883112 |
| <input type="checkbox"/>            | -            | sg-0f12c5a382a728fb7 | default                                 | yvc-07bf1d4263c8a31e4 | default VPC security group                 | 596059883112 |
| <input type="checkbox"/>            | -            | sg-04a8e657652cd7aa6 | default                                 | yvc-041d4a9e5de5dea2c | default VPC security group                 | 596059883112 |
| <input checked="" type="checkbox"/> | 3tier-alb-sg | sg-0edec15972673589e | tier3-web-app-ALBSecurityGroup-UVh...   | yvc-041d4a9e5de5dea2c | Security group for Application Load Bal... | 596059883112 |

tier3-app-asg

DetailsIntegrationsAutomatic scalingInstance managementInstance refreshSecurityMonitoringTags

Launch template

Launch template

 [lt-0d1de1cef70f5f86b](#)

tier3-app-template

Version

1

Description

-

[View details in the launch template console](#)

AMI ID

 [ami-0f3caa1cf4417e51b](#)

Security groups

-

Storage (volumes)

-

Instance type

t2.micro

Security group IDs

 [sg-09c143a7fd0ac1f96](#)

Key pair name

-

Owner

arn:aws:iam::596059883112:root

Create time

Tue Feb 24 2026 17:33:59 GMT+0530 (India Standard Time)

Request Spot Instances

No

Network

Availability Zones

use1-az2 (us-east-1b)  
use1-az1 (us-east-1a)

Subnet ID

 [subnet-0e039b28da877e3ca](#)  
 [subnet-0c3c6fa21cd6e8115](#)

Availability Zone distribution

Balanced best effort

Instance type requirements

Your Auto Scaling group adheres to the launch template for purchase option and instance type.

Health checks

Health check type

EC2, ELB

Health check grace period

300

tier3-app-db

Instance

Configuration

DB instance ID

tier3-app-db

Engine version

8.0.45

RDS Extended Support

Enabled

DB name

apddb

License model

General Public License

Option groups

[default:mysql-8-0](#)  In sync

Amazon Resource Name (ARN)

 [arn:aws:rds:us-east-1:596059883112:db:tier3-app-db](#)

Resource ID

db-WIDRAAWPND637GLXIVBLX54PM4

Created time

February 24, 2026, 17:23 (UTC+05:30)

DB instance parameter group

[default:mysql8.0](#)  In sync

Deletion protection

Disabled

Architecture settings

Non-multitenant architecture

Instance class

Instance class

db.t3.micro

vCPU

2

RAM

1 GB

Availability

Master username

admin

Master password

\*\*\*\*\*

IAM DB authentication

Not enabled

Multi-AZ

Yes

Secondary Zone

us-east-1b

Primary storage

Encryption

Not enabled

Storage type

General Purpose SSD (gp3)

Storage

20 GiB

Provisioned IOPS

3000 IOPS

Storage throughput

125 MiBps

Storage autoscaling

Disabled

Storage file system configuration

Current

Monitoring

Monitoring type

Database Insights - Standard

Performance Insights

Disabled

Enhanced Monitoring

Disabled

tier3-app-alb

tier3-app-alb

Actions

Details

Load balancer type

Application

Status

 Active

VPC

[vpc-041d4a9e5de5dea2c](#) 

Load balancer IP address type

IPv4

Scheme

Internet-facing

Hosted zone

Z355XDOTRQ7X7K

Availability Zones

[subnet-082ee9f82265b348d](#)  us-east-1a (use1-az1)  
[subnet-097489593d87d01c7](#)  us-east-1b (use1-az2)

Date created

February 24, 2026, 17:20 (UTC+05:30)

Load balancer ARN

 [arn:aws:elasticloadbalancing:us-east-1:596059883112:loadbalancer/app/tier3-app-alb/cf21eafaa15165cb](#)

DNS name

[tier3-app-alb-1508895037.us-east-1.elb.amazonaws.com](#) (A Record)

Listeners and rules

Network mappingResource mapSecurityMonitoringIntegrationsAttributesCapacityTags

Listeners and rules (1)

Filter listeners

Protocol:Port

Default action

Rules

ARN

Security policy

Default SSL/TLS certificate

mTLS

Trust store

 [HTTP:80](#)

Forward to target group

[tier3-app-tg](#)  1 (100%)  
Target group stickiness: Off

1 rule

 ARN

Not applicable

Not applicable

Not applicable

Not applicable

3-Tier Web Application

Stack tier3-web-app | CloLoad balancer details | ECAurora and RDS | us-east-

← → ↻ Not secure tier3-app-alb-1508895037.us-east-1.elb.amazonaws.com/index.php



## 3-Tier Web Application

[Home](#) | [View Users](#) | [Add User](#)



### Application Layer (This Instance)

Instance ID: i-02d83e3342000d50

Availability Zone: us-east-1b

Private IP: 10.0.12.98

Timestamp: 2026-02-24 12:13:16



### Database Layer (RDS)

 Database Status: **Connected Successfully**

Database Host: tier3-app-db-c43omiyq1000.us-east-1.rds.amazonaws.com

Database Name: appdb



### Application Statistics

Total Page Visits: 4

🏠 CloudKeeper 3-Tier Application - CloudFormation Deployment

Refresh this page to see load balancing in action!

Users - 3-Tier App

Stack tier3-web-app | CloLoad balancer details | ECAurora and RDS | us-east-

← → ↻ Not secure tier3-app-alb-1508895037.us-east-1.elb.amazonaws.com/users.php

## All Users

[Home](#) | [View Users](#) | [Add User](#)

| ID | Name         | Email                | Action                  |
|----|--------------|----------------------|-------------------------|
| 1  | Sirius Black | sirius@2001gmail.com | <button>Delete</button> |

Users - 3-Tier App

Stack tier3-web-app | CloLoad balancer details | ECAurora and RDS | us-east-

← → ↻ Not secure tier3-app-alb-1508895037.us-east-1.elb.amazonaws.com/users.php?deleted=1

## All Users

[Home](#) | [View Users](#) | [Add User](#)

User deleted successfully.

No users yet. [Add a user.](#)

## Add New User

[Home](#) | [View Users](#) | [Add User](#)

Name:

Email:

Please fill out this field.

Add User

## 4. You must deploy a same containerized ECS application in two region using CloudFormation StackSet, ensuring:

High availability (Multi-AZ)

Reusability

No hardcoded values

Production best practices

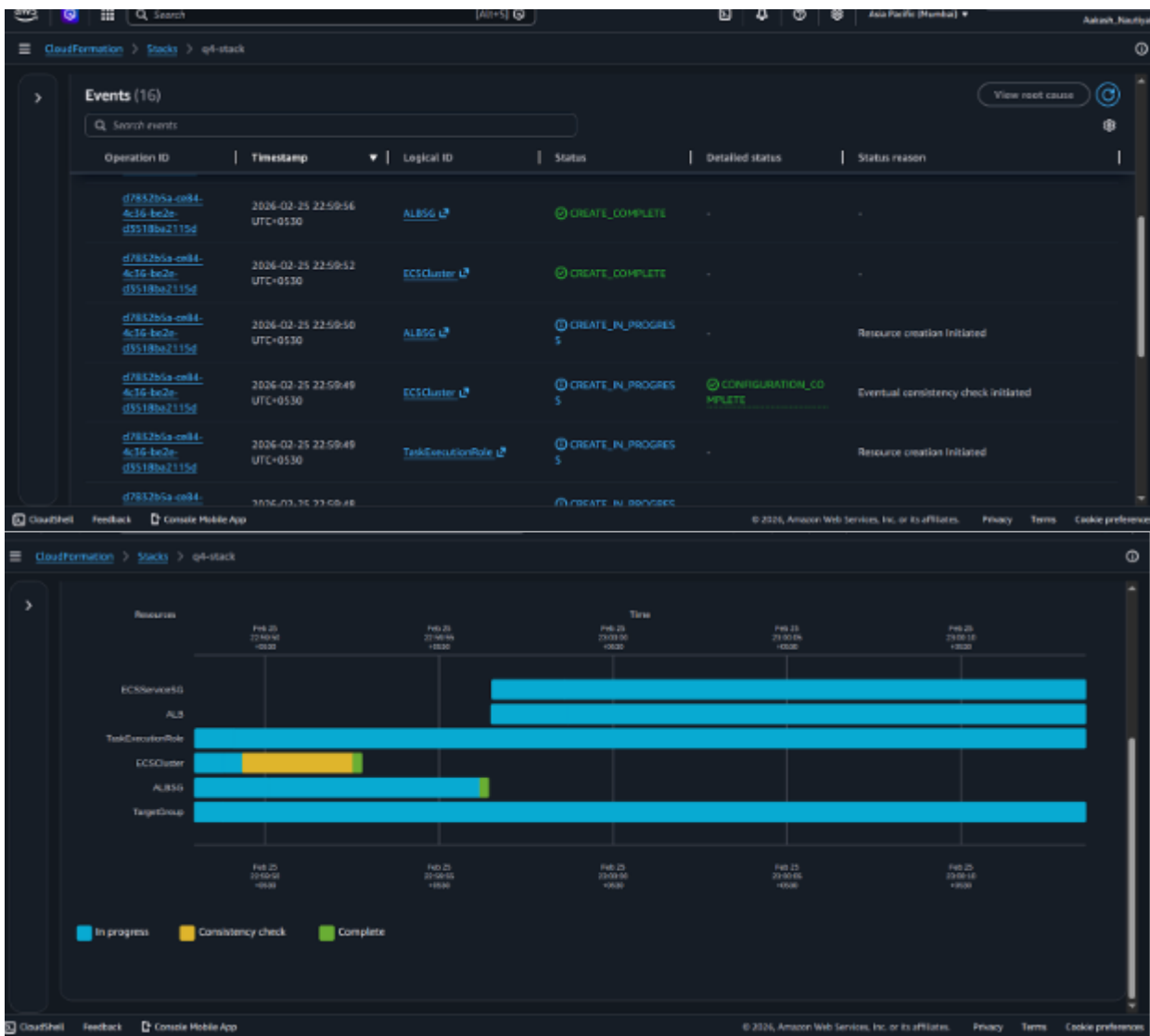
Ans:

The screenshot displays the AWS CloudFormation console. The top section shows the 'Create stack' wizard, which is currently on the 'Review and create' step. The parameters section includes:

- ContainerImage:** Container image URI (ECR or Docker Hub) with the value `deb / ubuntu/kub3 / 1.19.2-2`.
- ContainerPort:** 80
- SubnetIds:** At least two subnets in different AZs. The selected subnets are `subnet-04333fd8f3976a5d`, `subnet-05ed4adeff3402a71`, and `subnet-0ed3270ddff3e6480f`.
- VpcId:** Existing VPC ID with the value `vpc-073fd57cd17913781`.

The bottom section shows the 'Events (5)' page for the stack 'q4-stack'. The events table lists the following operations:

| Operation ID                         | Timestamp                    | Logical ID        | Status             | Detailed status | Status reason  |
|--------------------------------------|------------------------------|-------------------|--------------------|-----------------|----------------|
| d7852b5a-cm84-4c56-be2e-d5518ba2115d | -                            | -                 | -                  | -               | -              |
| d7852b5a-cm84-4c16-be2e-d5518ba2115d | 2026-02-25 22:59:47 UTC+0530 | TaskExecutionRole | CREATE_IN_PROGRESS | -               | -              |
| d7852b5a-cm84-4c16-be2e-d5518ba2115d | 2026-02-25 22:59:47 UTC+0530 | ECSCluster        | CREATE_IN_PROGRESS | -               | -              |
| d7852b5a-cm84-4c16-be2e-d5518ba2115d | 2026-02-25 22:59:47 UTC+0530 | ALB               | CREATE_IN_PROGRESS | -               | -              |
| d7852b5a-cm84-4c16-be2e-d5518ba2115d | 2026-02-25 22:59:47 UTC+0530 | TargetGroup       | CREATE_IN_PROGRESS | -               | -              |
| d7852b5a-cm84-4c16-be2e-d5518ba2115d | 2026-02-25 22:59:45          | q4-stack          | CREATE_IN_PROGRESS | -               | User Initiated |



## 5. Create an AWS CloudFormation template that deploys a cross-account IAM role. The role must:

Be assumable from a specified external AWS account.

Grant least-privilege access to EC2, RDS, and KMS.

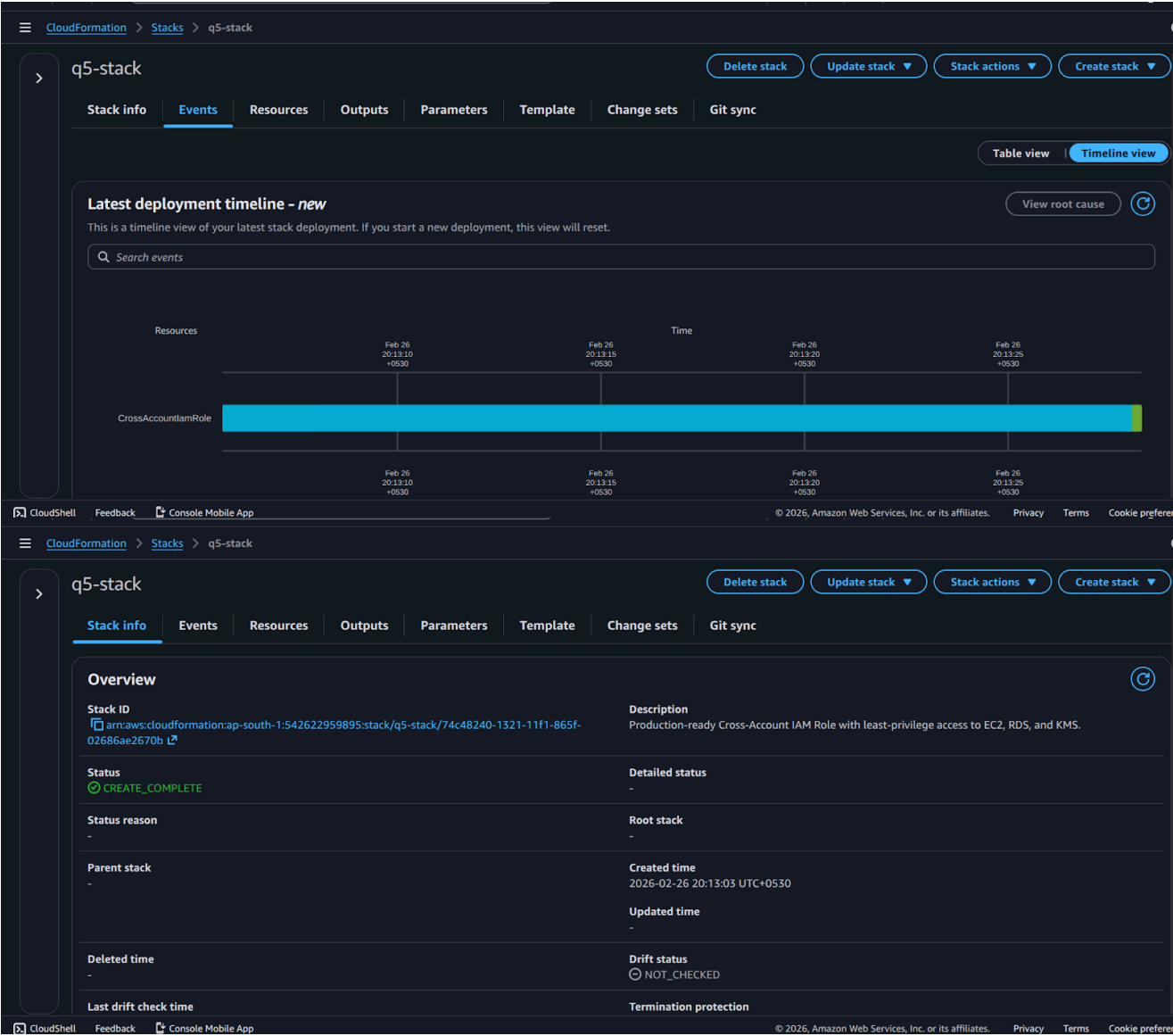
Include a proper trust policy allowing sts:AssumeRole from the external account.

Parameterize the external account ID.

Output the Role ARN after deployment.

Ensure the template follows security best practices and is production-ready.

Ans:



```
! Q5.yaml x
home > piyush > Desktop > ! Q5.yaml
1  AWSTemplateFormatVersion: '2010-09-09'
2  Description: >
3    Creates a cross-account IAM role that can be assumed by a specified external AWS account.
4    Grants least-privilege access to EC2, RDS, and KMS.
5
6  Parameters:
7    ExternalAccountId:
8      Type: String
9      Description: AWS Account ID allowed to assume this role (12-digit ID, no hyphens)
10     AllowedPattern: '^d{12}$'
11     ConstraintDescription: Must be a valid 12-digit AWS Account ID.
12
13   ExternalId:
14     Type: String
15     Description: External ID required to assume the role (recommended for third-party access)
16     NoEcho: true
17     MinLength: 8
18     MaxLength: 128
19
20   RoleName:
21     Type: String
22     Default: CrossAccountLeastPrivilegeRole
23     Description: Name of the IAM Role to create
24     MinLength: 3
25     MaxLength: 64
26
27  Resources:
28
29    CrossAccountRole:
30      Type: AWS::IAM::Role
31      Properties:
32        RoleName: !Ref RoleName
33        Description: Cross-account role with least privilege access to EC2, RDS, and KMS
34        MaxSessionDuration: 3600
35        AssumeRolePolicyDocument:
36          Version: '2012-10-17'
37          Statement:
38            - Sid: AllowExternalAccountAssumeRole
39              Effect: Allow
40              Principal:
41                AWS: !Sub arn:aws:iam::${ExternalAccountId}:root
42              Action: sts:AssumeRole
43              Condition:
44                StringEquals:
45                  sts:ExternalId: !Ref ExternalId
46
47        ManagedPolicyArns: []
48
49        Policies:
50          - PolicyName: LeastPrivilegeAccessPolicy
51            PolicyDocument:
52              Version: '2012-10-17'
53              Statement:
54
55                - Sid: EC2ReadOnlyAccess
56                  Effect: Allow
57                  Action:
58                    - ec2:DescribeInstances
59                    - ec2:DescribeVolumes
60                    - ec2:DescribeSnapshots
61                    - ec2:DescribeTags
62                    - ec2:DescribeSecurityGroups
63                    - ec2:DescribeVpcs
64                    - ec2:DescribeSubnets
65                  Resource: "*"
66
67                - Sid: RDSReadOnlyAccess
68                  Effect: Allow
69                  Action:
70                    - rds:DescribeDBInstances
71                    - rds:DescribeDBClusters
72                    - rds:DescribeDBSnapshots
73                    - rds:ListTagsForResource
74                  Resource: "*"
75
76                - Sid: KMSDecryptAccess
77                  Effect: Allow
78                  Action:
79                    - kms:Decrypt
80                    - kms:DescribeKey
81                  Resource: "*"
82                  Condition:
83                    StringEquals:
84                      kms:ViaService:
85                        - !Sub ec2:${AWS::Region}.amazonaws.com
86                        - !Sub rds:${AWS::Region}.amazonaws.com
```

```
Welcome  ! Q5.yaml x
home > piyush > Desktop > ! Q5.yaml
27  Resources:
29      CrossAccountRole:
31          Properties:
49          Policies:
90
91  Outputs:
92      CrossAccountRoleArn:
93          Description: ARN of the created cross-account IAM Role
94          Value: !GetAtt CrossAccountRole.Arn
95          Export:
96              Name: !Sub ${AWS::StackName}-CrossAccountRoleArn
```